

Department of Computer Science and Engineering
Academic Year 2021– 2022(Even Semester)

Degree, Semester & Branch: B.E, IV & CSE

Course Code & Title: CS8394 & Operating System

Name of the Faculty member (s): Dr. I Gethzi Ahila Poornima

Innovative Practice Description

- **Unit / Topic:** Unit I/ CPU Scheduling Algorithms
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO 2
- **Activity Chosen:** Active Learning- Think-Aloud Pair Problem-Solving (TAPPS)
- **Justification:**

The selected topic, "Problems using CPU Scheduling Algorithms," is one of the most fundamental and important concepts in operating systems. Because task scheduling by the operating system has a significant impact on CPU utilization, it is critical for students to understand this topic.

- **Time Allotted for the Activity:** 30 Minutes
- **Details of implementation**

Students can pair up to tackle case studies, complex problems, or understand text, with one person serving as the explainer and the other as the questioner. The explainers begin by stating the problem at hand before going into detail on how they would handle the case, problem, or interpretation. The questioners primarily listen, but they can also ask questions or provide useful recommendations. The students switch roles at a certain point, and this procedure continues until the activity is completed.

The chosen topics were covered in the previous class. Students were advised to prepare for the topic by solving a few related questions. The previous classes covered CPU scheduling techniques (both preemptive and non-preemptive), such as FCFS, SRF, RR, and priority scheduling strategies. They were instructed to establish a pair on the day of the exercise depending on their own preferences. A problem in CPU scheduling techniques is assigned to each pair. They were given five different problems in all. (One for each pair.)

They were given 20 minutes to talk on the issue at hand. The explainer will be one of the members of the pair. He or she will talk to the other about the steps required in solving the problem. The other member (questioner) will next ask the explanation to clarify his doubt. The members' positions were switched about, and the process continues. Two pairs were chosen at random after 20 minutes to present the same on stage. Figure 1 displays a few sample of photos taken during the class session.

- **CO – PO / PSO mapping:**

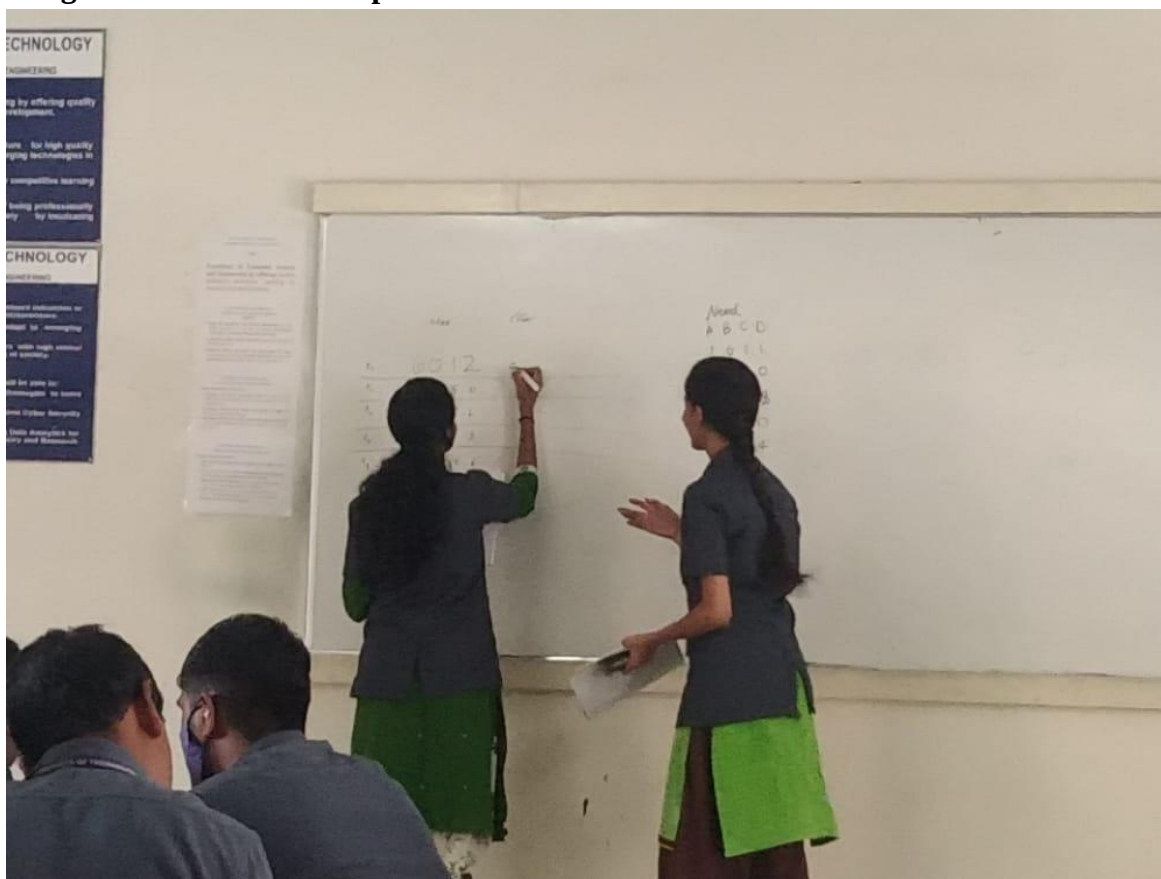
CO	PO2	PO3	PSO1
CO5	1	2	2

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:

Innovative practice	PO1	PO4	PSO1
	2	2	2
Justification for correlation	Students could recognize the importance of CPU scheduling, which improved CPU usage and reduced average turnaround time and waiting time. They were also able to select the appropriate scheduling algorithms based on the requirements.	Students will be able to analyze the functions of Operating system based on the knowledge gained through the concept of scheduling algorithms.	The basic understanding of the operating system functionalities such as CPU scheduling will enable the students to develop software to solve the complex computation tasks..

- Images / Screenshot of the practice:



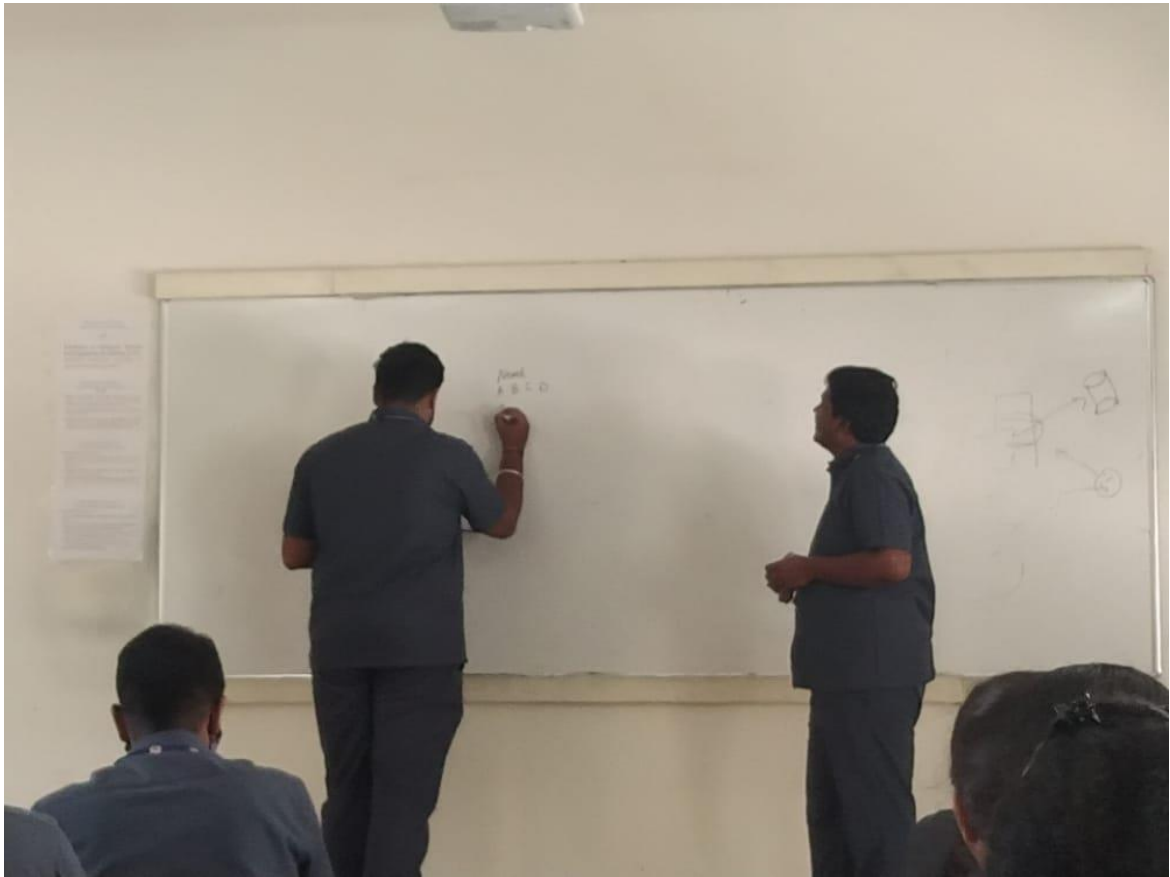


Figure 1: Sample sheet & picture taken

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Most of the students felt that this event helped them to analyze and compare the performance of various CPU scheduling algorithms in terms of waiting time and turnaround time.

- ❖ ***Benefit of the practice:***

- All of the students were engaged in the activities and had a realistic experience of discussion about the problem.
- They believed they could compare the performance of various scheduling algorithms.

- ❖ ***Challenges faced in implementation:***

- Few students hesitated to do the activity on stage.

Signature of Faculty Member

HOD